The Point-Source First Law Fails in 1d, Where CHM Is a Theorem

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Abstract

Paper 35 found that a 3d point source makes δS track enclosed energy, not the CHM weight $R^2 - r^2$. That is a 3d statement only if the same operators recover CHM on a 1d massless interval, where Casini–Huerta–Myers is a theorem for K.

They do not. $N = 200$, $L = 16$, 185 intervals, ε on one site. δS is linear in ε (Pearson 1). $\rho(\delta S, P_{\mathrm{CHM}}) = -0.054$. $\rho(\delta S, P_{\mathrm{flat}}) = -0.276$. Residuals: CHM 0.999, flat 0.961. C2 FAIL. C4 FAIL.

The exact modular identity holds: $\rho(\delta S, \mathrm{Tr}(K_{\mathrm{mid}}\Delta C)) = 1$ (relative error 0.14%). The vacuum first law does not: $\rho(\delta S, \mathrm{Tr}(K_{\mathrm{vac}}\Delta C)) = -0.025$.

Auditor, $N = 160$, $L = 12$: C2 REFUTED; K_{vac} REFUTED; K_{mid} CONFIRMED.

Paper 35 is not a 3d no-go. The instrument tests a vacuum first-law proxy that already fails the theorem case. Not $8\pi G$. Not de Sitter.

1 What this is

CHM still describes the *shape* of K (Paper 25). This note asks whether δS of a sliding interval, after a point Hamiltonian source, equals a local functional of δe. That is the Paper 35 comparison, moved to 1d.

It fails because $\delta S = \mathrm{Tr}(K_{\mathrm{mid}}\Delta C)$, not $\mathrm{Tr}(K_{\mathrm{vac}}\Delta C)$. A point potential is not a soft state variation at fixed modular Hamiltonian.

Admission. *I ran the calibration that can kill the 3d reading of Paper 35. It killed it. I did not write that CHM is false, and I did not write Einstein.*